Final C2O Strategy Report

C2O Coursework Submission

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Executive Summary

The promoted final strategy is `phase2_g5_05_expanding`: a daily, dollar-neutral, close-to-open long-short strategy using the volatility-scaled overnight-return target with expanding-window transparent feature-weight estimation. The target is:

```
target = overnight_next / (vol20 / sqrt(252))
```

`overnight_next` is the next close-to-open return, and `vol20` is a trailing 20-day close-to-close volatility estimate shifted by one trading day. The feature set, basket rule, weighting, raw-score ranking, tiered borrow-cost treatment, transaction costs, 5% ADV20 participation cap, close-to-open execution, and development cutoff remain unchanged from the previous final implementation.

At the main headline AUM of 250M, the promoted final strategy earns `7.31%` net annual return, `4.97%` net volatility, net Sharpe `1.445`, and maximum drawdown `-10.19%` over 2010-2024. The previous 4-year rolling champion had 250M net Sharpe `0.811`; that value is now only a comparison point, not the final strategy.

The controlled Phase 2 screen supports promotion: validation 2019-2022 250M Sharpe improves to `2.279`, internal holdout 2023-2024 remains positive at `0.620`, and full 2010-2024 Sharpe is `1.445`. The holdout result is much lower than validation, so the report treats this as controlled evidence of improvement, not a promise of live performance.

2025-2026 remains held out for marker evaluation. No 2025+ data is used in development outputs.

Final Strategy Configuration

- Target: `volatility_scaled_overnight_return`.
- Target definition: `overnight_next / (vol20 / sqrt(252))`.
- Training window: expanding. For each scored year Y, training starts at 2010-01-01 and ends before year Y begins.
- Alpha learning: 50% prior / 50% learned transparent correlation weights.
- Basket: top/bottom 3% of eligible names, with minimum 15 names per side.
- Weighting: equal weight within each side, subject to capacity caps.
- Ranking: raw alpha score.
- Short treatment: tiered borrow cost only; no hard short exclusion.
- Execution: enter at day-t closing auction and exit at day-t+1 opening auction.
- Development cutoff: 2024-12-31.

Volatility scaling makes the training label risk-adjusted. Expanding-window training uses more historical overnight evidence as time progresses, which can stabilise transparent feature-weight estimation relative to a fixed 4-year rolling window. The performance improvement is not from cost relaxation, borrow relaxation, participation-cap relaxation, execution-timing changes, external data, intraday data, or new features.

Headline Performance

AUM	Net annual return	Net vol	Net Sharpe	Max drawdown	Gross Sharpe	Avg gross exposure	Cap-binding days
50M	6.66%	5.03%	1.308	-13.82%	3.539	99.92%	3547
250M	7.31%	4.97%	1.445	-10.19%	3.112	74.97%	3773
1B	3.50%	2.69%	1.293	-5.56%	2.321	25.35%	3773

250M is the main headline AUM because it balances realised gross exposure, capacity use, and cost drag. 1B is included as a capacity boundary case.

Champion-Challenger Evidence

Metric	Previous champion	Promoted final
Validation 2019-2022 250M net Sharpe	1.107	2.279
Internal holdout 2023-2024 250M net Sharpe	0.003	0.620
Full 2010-2024 250M net Sharpe	0.811	1.445
Full 2010-2024 max drawdown	-10.19%	-10.19%
Full 2010-2024 worst 12m return	-10.09%	-10.09%

The selected challenger passes the promotion audit. It improves validation Sharpe, remains positive in 2023-2024 internal holdout, improves full-period Sharpe, and does not worsen full-period maximum drawdown. The previous 0.811 Sharpe strategy is archived under `outputs/previous_champion_archive_after_phase2/` and `outputs/current_champion_archive/`.

The 2023-2024 holdout Sharpe of `0.620` is materially lower than the validation Sharpe of `2.279`. This degradation is expected in a controlled screen and should be disclosed plainly.

Costs And Execution

The cost model is unchanged:

- Commission: 0.5 bps per leg.
- Auction slippage: 1.5 bps per leg.
- Total non-borrow round-trip cost: 4.0 bps.
- Borrow Tier A: 40 bps p.a.
- Borrow Tier B: 200 bps p.a.
- Borrow Tier C: 800 bps p.a.
- Borrow daily charge: annual rate / 252.
- Participation cap: 5% ADV20.

Sharpe drag values below are Sharpe-unit drags, not percentages.

AUM	Gross Sharpe	Commission drag	Slippage drag	Borrow drag	Net Sharpe	Avg daily borrow bps
50M	3.539	0.501	1.503	0.227	1.308	0.451
250M	3.112	0.378	1.137	0.152	1.445	0.301
1B	2.321	0.233	0.704	0.091	1.293	0.098

At 250M, gross Sharpe is `3.112` and net Sharpe is `1.445` after commission, auction slippage, and borrow costs. Auction slippage remains the largest explicit cost drag.

Capacity

AUM	Avg gross exposure	Cap-binding days	Fraction cap-binding days	Avg participation	Position days at cap
50M	99.92%	3547	93.99%	2.04%	16.16%
250M	74.97%	3773	99.97%	4.02%	66.29%
1B	25.35%	3773	99.97%	4.29%	75.40%

At 250M, average gross exposure used is `74.97%`. At 1B it falls to `25.35%`, which means the strategy cannot always deploy target risk under the fixed 5% ADV20 cap. This is an honest capacity result rather than an implementation error.

Borrow Treatment

The strategy uses provided point-in-time short-interest proxies plus a one-trading-day decision lag to assign borrow tiers. Borrow affects net returns through the explicit daily charge; it does not change the raw short selection in the promoted final strategy.

Tier	Annual rate	Daily charge
A	40 bps p.a.	annual rate / 252
B	200 bps p.a.	annual rate / 252
C	800 bps p.a.	annual rate / 252

The report does not claim independent prime-broker borrow-fee validation. No direct securities-lending or locate-rate data is used.

Feature Ablation Interpretation

Variant	Removed group	IC mean	IC t-stat	Net annual return	Net vol	Net Sharpe	Max drawdown
full_model	none	0.029	9.934	7.31%	4.97%	1.445	-10.19%
remove_return/reversal	return/reversal	0.011	3.333	-3.62%	3.26%	-1.114	-46.66%
remove_risk/liquidity/size	risk/liquidity/size	0.028	10.863	5.89%	5.12%	1.144	-8.24%
remove_fundamental/value/quality	fundamental/value/quality	0.03	10.203	7.01%	4.90%	1.407	-9.75%
remove_earnings/revision	earnings/revision	0.028	9.754	7.31%	4.86%	1.477	-10.19%
remove_short-interest/borrow-stress	short-interest/borrow-stress	0.029	9.846	7.70%	5.00%	1.508	-11.32%
remove_industry-return	industry-return	0.029	9.942	7.27%	4.96%	1.442	-10.19%

The ablation is a dependence audit, not a retuning exercise. Return/reversal is the clearest alpha contributor: removing it lowers 250M Sharpe from `1.445` to `-1.114`. Some other removals improve Sharpe, especially short-interest/borrow-stress and earnings/revision in this frozen no-retuning table. That should not be hidden. It means some variables may be acting as risk, capacity, crowding, turnover, or cost-control information rather than standalone alpha predictors.

Robustness

Year-By-Year 250M Results

Year	Return	Vol	Sharpe	Max drawdown	Avg gross exposure
2010	-10.09%	3.42%	-3.094	-10.19%	42.46%
2011	21.39%	5.63%	3.474	-2.08%	67.54%
2012	6.70%	2.80%	2.333	-1.68%	56.93%
2013	5.61%	2.38%	2.307	-1.17%	64.02%
2014	2.57%	3.66%	0.712	-5.01%	80.35%
2015	3.91%	2.46%	1.575	-1.06%	64.59%
2016	3.30%	2.33%	1.41	-1.53%	47.44%
2017	1.64%	2.05%	0.804	-1.35%	61.08%
2018	3.08%	3.84%	0.808	-3.38%	81.30%
2019	12.30%	3.64%	3.206	-1.03%	80.89%
2020	15.53%	9.57%	1.557	-4.78%	88.45%
2021	10.83%	5.96%	1.755	-5.47%	96.18%
2022	32.24%	8.02%	3.525	-2.63%	99.23%
2023	9.70%	5.32%	1.768	-4.85%	96.07%
2024	-2.61%	5.73%	-0.434	-5.09%	98.07%

Pre/Post Split

Subperiod	Return	Vol	Sharpe	Max drawdown	Avg gross exposure
2010_2016	4.42%	3.45%	1.273	-10.19%	60.48%
2017_2024	9.90%	5.99%	1.608	-7.64%	87.66%

Stress Windows

Window	Return	Vol	Sharpe	Max drawdown	Avg gross exposure
late_2018	-2.35%	4.71%	-0.482	-1.92%	85.28%
2020_q1	34.41%	10.78%	2.799	-4.14%	88.95%
2022_drawdown	32.24%	8.02%	3.525	-2.63%	99.23%

Tail And Serial Correlation

At 250M, worst 3-month return is `-4.46%`, worst 6-month return is `-7.31%`, and worst 12-month return is `-10.09%`. The top 5% best days contribute `1.461` times total PnL, which means daily PnL is meaningfully concentrated. Lag-1 return autocorrelation is `-0.007`, so there is no large positive serial-correlation effect inflating annualised Sharpe in the final output.

Point-In-Time Controls

The promotion audit passes the following controls:

- No 2025+ data is used; promoted cache max date is `2024-12-31`.
- The target uses the future overnight return only as a historical training label.
- `vol20` is trailing and shifted before use in the target denominator.
- For each scored year, expanding-window training uses only dates strictly before that year.
- Cross-sectional ranks use only the same-day available cross-section.
- Earnings filtering uses `strat_trading_date` and respects AMC/BMO timing.
- Short-interest fields use provided point-in-time proxies from the coursework data package plus a one-trading-day decision lag.
- No day-t close/high/low enters a decision-time feature unless shifted appropriately.

Baseline And Previous Champion

The original baseline is preserved in `outputs/baseline_archive/`. Its original 250M net Sharpe was approximately `0.374`; the logged experiment baseline was approximately `0.379`. The previous 4-year rolling champion is preserved in `outputs/previous_champion_archive_after_phase2/` with 250M Sharpe `0.811`.

The promoted final strategy should be compared against those archives without relabelling them as final.

Reproduction

The final outputs are regenerated with:

```
PYTHON=/opt/anaconda3/bin/python3 make reproduce
PYTHON=/opt/anaconda3/bin/python3 make test
```

The latest test run passed with `13 passed`. The submitted package includes final daily returns, positions for 50M/250M/1B, QuantStats HTML, report-ready audit evidence, and archive notes.

Limitations

The main limitations are straightforward: no external prime-broker borrow validation, no independent raw FINRA short-interest publication-date reconstruction, no separate market-impact model beyond the fixed brief cost schedule and realised participation reporting, and visible capacity constraints at 1B. The 2025-2026 period remains held out for marker evaluation.

Final Decision

Promotion passed. The final strategy is the volatility-scaled overnight-return target with expanding-window weight estimation: `phase2_g5_05_expanding`. The previous 0.811 champion remains preserved for audit comparison, while the live final headline is the promoted 250M result: `7.31%` net annual return, `4.97%` net volatility, Sharpe `1.445`, and max drawdown `-10.19%`.